Analysis of Stochastic Gradient Descent

Materials from *Gartner, B., He, N., and Jaggi, M. Lectures notes on Optimization for Data Science.*

May 17, 2026

Stochastic Optimization

General form:

$$\min_{\mathbf{x} \in \mathsf{C}} F(x) = \mathbb{E}_{\boldsymbol{\xi}}[f(\mathbf{x}, \boldsymbol{\xi})] \qquad \text{(Stochastic Optimization)}$$

where $f(\mathbf{x}, \boldsymbol{\xi})$ is a function involving the decision variable $\mathbf{x}$ and a random variable (vector) $\boldsymbol{\xi} \sim \mathbb{P}(\boldsymbol{\xi})$.

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In particular, given a set of function $\{f_i(\mathbf{x})\}_{i=1}^{n}$ and let $\boldsymbol{\xi} \sim \texttt{Unif}\{1, \ldots, n\}$. Define $f(\mathbf{x}, \boldsymbol{\xi}) = f_{\boldsymbol{\xi}}(\mathbf{x})$, then

$$\min_{\mathbf{x}\in \mathsf{C}} \; F(\mathbf{x}) = \mathbb{E}_{\boldsymbol{\xi}}[f(\mathbf{x}, \boldsymbol{\xi})] := \frac{1}{n}\sum_{i=1}^{n} f_i(\mathbf{x}) \qquad \text{(Finte-Sum Problem)}$$

Stochastic Gradient Descent

Assume that $f(\mathbf{x}, \boldsymbol{\xi})$ is continuously differentiable for any instance of $\boldsymbol{\xi}$. The Stochastic Gradient Descent (SGD) updates the numerical solution by

$$\mathbf{x}_{t+1} = \Pi_{\mathsf{C}}\left(\mathbf{x}_t - \gamma_t \nabla f(\mathbf{x}_t, \boldsymbol{\xi}_t)\right),$$

where $\boldsymbol{\xi}_t \sim \mathbb{P}(\boldsymbol{\xi})$ and ∇ is taken over the argument $\mathbf{x}$, starting from a deterministic initialization $\mathbf{x}_1$.

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The corresponding update formula of solving (Finte-Sum Problem) by SGD is written as

$$\mathbf{x}_{t+1} = \Pi_{\mathsf{C}} \left(\mathbf{x}_t - \gamma_t \nabla f_{i_t}(\mathbf{x}_t) \right), \quad i_t \sim \texttt{Unif}\{1, \dots, n\}.$$

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Note that

$$\mathbb{E}[\nabla f_{i_t}(\mathbf{x}_t) | \mathbf{x}_t] = \sum_{i=1}^{n} \nabla f_i(\mathbf{x}_t) \cdot \mathbb{P}(i_t = i) = \frac{1}{n} \sum_{i=1}^{n} \nabla f_i(\mathbf{x}_t) = \nabla F(\mathbf{x}_t)$$

Stochastic Gradient Descent

In our analysis, we always assume that

$$\mathbb{E}[\nabla f(\mathbf{x}, \boldsymbol{\xi})] = \int \nabla f(\mathbf{x}, \boldsymbol{\xi}) \mathbb{P}(\boldsymbol{\xi}) d\boldsymbol{\xi} = \nabla \int f(\mathbf{x}, \boldsymbol{\xi}) \mathbb{P}(\boldsymbol{\xi}) d\boldsymbol{\xi} = \nabla F(\mathbf{x}).$$

Stochastic Gradient Descent

Example

For ordinary least squares regression,

$$\min_{\mathbf{x}\in\mathbb{R}^d} \; F(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{a}_i^T\mathbf{x} - b_i)^2, \quad f_i(\mathbf{x}) = (\mathbf{a}_i^T\mathbf{x} - b_i)^2$$

where $\{(\mathbf{a}_i, b_i)\}_{i=1}^n$ is the set of training samples.

Stochastic Gradient Descent

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$$\texttt{Gradient Descent (GD)}: \quad \mathbf{x}_{t+1} = \mathbf{x}_t - 2\gamma_t \sum_{i=1}^{n}(\mathbf{a}_i^T\mathbf{x} - b_i)\mathbf{a}_i$$

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- **GD: n samples are used in per-iteration.**
- **SGD: one sample is used in per-iteration.**

Stochastic Gradient Descent

$$\mathbf{x}_{t+1} = \Pi_{\mathsf{C}}\left(\mathbf{x}_t - \gamma_t \nabla f(\mathbf{x}_t, \boldsymbol{\xi}_t)\right)$$

- Since $\nabla f(\mathbf{x}_t, \boldsymbol{\xi}_t)$ is a random variable, we cannot guarantee that $\nabla f(\mathbf{x}_t, \boldsymbol{\xi}_t) \to 0$ as $t \to \infty$. Therefore, $\gamma_t \to 0$ as $t \to \infty$ is needed to ensure convergence.
- $\{\mathbf{x}_t\}_{t \geq 1}$ is a random process of which randomness originates from $\boldsymbol{\xi}_t$, and so is $\{F(\mathbf{x}_t)\}_{t \geq 1}$.

Convergence for Strongly Convex Functions

Theorem 1 (Convergent Step Size)

Assume that $F(\mathbf{x})$ is μ-strongly convex, and $\exists M > 0$, s.t. $\mathbb{E}[\|\nabla f(\mathbf{x}, \boldsymbol{\xi})\|_2^2] \leq M^2$, $\forall \mathbf{x} \in$ C, then SGD with $\gamma_t = \frac{\gamma}{t}$ at iteration t where $\gamma > 1/(2\mu)$ satisfies

$$\mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right] \leq \frac{C(\gamma)}{t}, \quad \text{where } C(\gamma) = \frac{\gamma^2 M^2}{2\mu\gamma - 1}$$

Proof of Theorem 1.

Let $\mathbf{g}_{\boldsymbol{\xi}_t} = \nabla f(\mathbf{x}_t, \boldsymbol{\xi}_t)$. Given $\mathbf{x}_t$, by the non-expansive property of the projection operator, we have

$$\begin{aligned}\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 &= \|\Pi_{\mathsf{C}}(\mathbf{x}_t - \gamma_t \mathbf{g}_{\boldsymbol{\xi}_t}) - \mathbf{x}^*\|^2 \\ &\leq \|\mathbf{x}_t - \mathbf{x}^* - \gamma_t \mathbf{g}_{\boldsymbol{\xi}_t}\|^2 \\ &= \|\mathbf{x}_t - \mathbf{x}^*\|^2 - 2\gamma_t \langle \mathbf{g}_{\boldsymbol{\xi}_t}, \mathbf{x}_t - \mathbf{x}^* \rangle + \gamma_t^2 \|\mathbf{g}_{\boldsymbol{\xi}_t}\|^2 \quad \text{a.s.}\end{aligned}$$

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Taking expectation both sides conditioned on $\mathbf{x}_t$ yields

$$\mathbb{E}[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 | \mathbf{x}_t] \leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 - 2\gamma_t \langle \nabla F(\mathbf{x}_t), \mathbf{x}_t - \mathbf{x}^* \rangle + \gamma_t^2 M^2$$

□

Proof of Theorem 1.

By strong convexity of $F(\mathbf{x})$, we have

$$\langle \nabla F(\mathbf{x}_t), \mathbf{x}_t - \mathbf{x}^* \rangle \geq F(\mathbf{x}_t) - F(\mathbf{x}^*) + \frac{\mu}{2}\|\mathbf{x}_t - \mathbf{x}^*\|^2,$$

and

$$\begin{aligned} F(\mathbf{x}_t) - F(\mathbf{x}^*) &\geq \langle \nabla F(\mathbf{x}^*), \mathbf{x}_t - \mathbf{x}^* \rangle + \frac{\mu}{2}\|\mathbf{x}_t - \mathbf{x}^*\|^2 \\ &\geq \frac{\mu}{2}\|\mathbf{x}_t - \mathbf{x}^*\|^2 \end{aligned}$$

where the last inequality comes from the optimality of $\mathbf{x}^*$.

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and

$$\begin{aligned} F(\mathbf{x}_t) - F(\mathbf{x}^*) &\geq \langle \nabla F(\mathbf{x}^*), \mathbf{x}_t - \mathbf{x}^* \rangle + \frac{\mu}{2} \|\mathbf{x}_t - \mathbf{x}^*\|^2 \\ &\geq \frac{\mu}{2} \|\mathbf{x}_t - \mathbf{x}^*\|^2 \end{aligned}$$

where the last inequality comes from the optimality of $\mathbf{x}^*$.

Putting together, we have

$$\mathbb{E}[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 | \mathbf{x}_t] \leq (1 - 2\mu\gamma_t) \|\mathbf{x}_t - \mathbf{x}^*\|^2 + \gamma_t^2 M^2.$$

$\square$

Proof of Theorem 1.

And by the tower formula, we get

$$\begin{aligned}\mathbb{E}[\|\mathbf{x}_{t+1}-\mathbf{x}^*\|^2] &\leq (1-2\mu\gamma_t)\mathbb{E}[\|\mathbf{x}_t-\mathbf{x}^*\|^2]+\gamma_t^2M^2 \\ &\leq (1-\frac{2\mu\gamma}{t})\mathbb{E}[\|\mathbf{x}_t-\mathbf{x}^*\|^2]+\frac{\gamma^2M^2}{t^2}.\end{aligned}$$

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Let $t = 1$,

$$0 \leq \mathbb{E}[\|\mathbf{x}_2 - \mathbf{x}^*\|^2] \leq (1 - 2\mu\gamma)\|\mathbf{x}_1 - \mathbf{x}^*\|^2 + \gamma^2 M^2.$$

Since $\gamma > 1/(2\mu)$,

$$\|\mathbf{x}_1 - \mathbf{x}^*\|^2 \leq \frac{\gamma^2 M^2}{2\mu\gamma - 1} = \frac{C(\gamma)}{1}.$$

We complete the proof by induction. $\square$

Convergence for Strongly Convex Functions

Theorem 2 (Constant Step Size)

Assume that $F(\mathbf{x})$ is both μ-strongly convex and L-smooth. Moreover, assume that stochastic gradient satisfies that

$$\mathbb{E}\left[\|\nabla f(\mathbf{x}, \boldsymbol{\xi})\|_2^2\right] \leq \sigma^2 + c\|\nabla F(\mathbf{x})\|^2.$$

Then, SGD with constant stepsize $\gamma_t \equiv \gamma \leq \frac{1}{cL}$ achieves:

$$\mathbb{E}[F(\mathbf{x}_t) - F(\mathbf{x}^*)] \leq \frac{\gamma L \sigma^2}{2\mu} + (1 - \gamma\mu)^{t-1} \left(F(\mathbf{x}_1) - F(\mathbf{x}^*)\right),$$

where $\mathbf{x}^*$ is the optimal solution and $\mathsf{C} = \mathbb{R}^d$.

- $\lim\limits_{t\to\infty} \left(F(\mathbf{x}_t) - F(\mathbf{x}^*)\right) \leq \frac{\gamma L \sigma^2}{2\mu}$
- **What if $\mathsf{C} \subset \mathbb{R}^d$?**

Proof of Theorem 2.

Since $\mathsf{C} = \mathbb{R}^d$, given $\mathbf{x}_t$ and let $\mathbf{g}_{\boldsymbol{\xi}_t} = \nabla f(\mathbf{x}_t, \boldsymbol{\xi}_t)$, we have

$$\mathbf{x}_{t+1} = \Pi_{\mathsf{C}}(\mathbf{x}_t - \gamma \mathbf{g}_{\boldsymbol{\xi}_t}) = \mathbf{x}_t - \gamma \mathbf{g}_{\boldsymbol{\xi}}, \quad \boldsymbol{\xi}_t \sim \mathtt{Unif}\{1, \dots, n\}.$$

And the L-smoothness of $F(\mathbf{x})$ yields

$$\begin{aligned} F(\mathbf{x}_{t+1}) - F(\mathbf{x}_t) &\leq \langle \nabla F(\mathbf{x}_t), \mathbf{x}_{t+1} - \mathbf{x}_t \rangle + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\ &= -\gamma \langle \nabla F(\mathbf{x}_t), \mathbf{g}_{\boldsymbol{\xi}_t} \rangle + \frac{L\gamma^2}{2} \|\mathbf{g}_{\boldsymbol{\xi}_t}\|^2 \quad \mathtt{a.s.} \end{aligned}$$

Proof of Theorem 2.

Since $\mathsf{C} = \mathbb{R}^d$, given $\mathbf{x}_t$ and let $\mathbf{g}_{\boldsymbol{\xi}_t} = \nabla f(\mathbf{x}_t, \boldsymbol{\xi}_t)$, we have

$$\mathbf{x}_{t+1} = \Pi_{\mathsf{C}}(\mathbf{x}_t - \gamma \mathbf{g}_{\boldsymbol{\xi}_t}) = \mathbf{x}_t - \gamma \mathbf{g}_{\boldsymbol{\xi}}, \quad \boldsymbol{\xi}_t \sim \texttt{Unif}\{1, \ldots, n\}.$$

And the L-smoothness of $F(\mathbf{x})$ yields

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Taking expectation both sides conditioned on $\mathbf{x}_t$ yields

$$\mathbb{E}[F(\mathbf{x}_{t+1}) - F(\mathbf{x}_t)] \leq -\gamma \langle \nabla F(\mathbf{x}_t), \mathbb{E}[\mathbf{g}_{\boldsymbol{\xi}_t}] \rangle + \frac{L\gamma^2}{2} \mathbb{E}[\|\mathbf{g}_{\boldsymbol{\xi}_t}\|^2].$$

□

Proof of Theorem 2.

By the assumption and $\mathbb{E}[\mathbf{g}_{\boldsymbol{\xi}_t}] = \nabla F(\mathbf{x}_t)$, we get

$$\mathbb{E}[F(\mathbf{x}_{t+1}) - F(\mathbf{x}_t)] \leq -\gamma \left(1 - \frac{c\gamma L}{2}\right) \|\nabla F(\mathbf{x}_t)\|^2 + \frac{\gamma^2 L \sigma^2}{2}. \tag{$\spadesuit$}$$

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Since $F(x)$ is μ-strongly convex,

$$\frac{1}{2\mu}\|F(\mathbf{x}_t)\|^2 \geq F(\mathbf{x}_t) - F(\mathbf{x}^*) \quad \text{a.s.} \qquad (\diamondsuit)$$

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Define $\Delta_t = F(\mathbf{x}_t) - F(\mathbf{x}_*)$. Note that $\gamma \leq \frac{1}{cL}$ and hence $1 - \frac{\gamma c L}{2} \geq 0$.

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Define $\Delta_t = F(\mathbf{x}_t) - F(\mathbf{x}_*)$. Note that $\gamma \leq \frac{1}{cL}$ and hence $1 - \frac{\gamma c L}{2} \geq 0$. Combing ($\spadesuit$) and ($\diamondsuit$) yields

$$\begin{aligned}
\mathbb{E}[\Delta_{t+1}] &\leq \left(1 - 2\gamma\mu + \gamma^2 c\mu L\right) \mathbb{E}[\Delta_t] + \frac{\gamma^2 L \sigma^2}{2} \\
&\leq (1 - \gamma\mu)\, \mathbb{E}[\Delta_t] + \frac{\gamma^2 L \sigma^2}{2}.
\end{aligned}$$

$\square$

Proof of Theorem 2.

The above inequality can be rearranged as

$$\mathbb{E}[\Delta_{t+1}] - \frac{\gamma L \sigma^2}{2\mu} \leq (1 - \gamma\mu) \left[\mathbb{E}[\Delta_t] - \frac{\gamma L \sigma^2}{2\mu} \right]$$

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\begin{aligned}
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&\leq (1 - \gamma\mu)^t \left[\mathbb{E}[\Delta_1] - \frac{\gamma L \sigma^2}{2\mu} \right] \\
&\leq (1 - \gamma\mu)^t \, \mathbb{E}[\Delta_1] \quad \forall t \geq 1
\end{aligned}
$$

$\square$

Convergence for Nonconvex Functions

Theorem 3

Assume that $F(\mathbf{x})$ is L-smooth and the stochastic gradient has bounded variance, namely, $\mathbb{E}\|\nabla f(\mathbf{x}, \boldsymbol{\xi}) - \nabla F(\mathbf{x})\|^2 \le \sigma^2$. Then SGD with stepsize $\gamma_t \equiv \gamma := \min\{\frac{1}{L}, \frac{\gamma_0}{\sigma\sqrt{T}}\}$ achieves

$$\mathbb{E}\left[\|\nabla F(\hat{\mathbf{x}}_T)\|^2\right] \le \frac{2L(F(\mathbf{x}_1) - F(\mathbf{x}^*))}{T} + \frac{\sigma}{\sqrt{T}}\left(\frac{2(F(\mathbf{x}_1) - F(\mathbf{x}^*))}{\gamma_0} + L\gamma_0\right)$$

where $\hat{\mathbf{x}}_T$ is selected uniformly at random from $\{\mathbf{x}_1, \ldots, \mathbf{x}_T\}$ and $\mathsf{C} = \mathbb{R}^d$.

Proof of Theorem 3.

From the bounded variance of the stochastic gradient, we have

$$\mathbb{E}[\|\nabla f(\mathbf{x}, \boldsymbol{\xi})\|^2] \leq \sigma^2 + \|\nabla F(\mathbf{x})\|^2$$

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Given $\mathbf{x}_t$ and let $\mathbf{g}_{\boldsymbol{\xi}_t} = \nabla f(\mathbf{x}_t, \boldsymbol{\xi}_t)$, following the proof of Theorem 2 yields

$$\mathbb{E}[F(\mathbf{x}_{t+1}) - F(\mathbf{x}_t)] \leq -\gamma \left(1 - \frac{\gamma L}{2}\right) \|\nabla F(\mathbf{x}_t)\|^2 + \frac{\gamma^2 L \sigma^2}{2},$$

which can be rearranged as

$$\gamma \left(1 - \frac{\gamma L}{2}\right) \|\nabla F(\mathbf{x}_t)\|^2 \leq \mathbb{E}[F(\mathbf{x}_t) - F(\mathbf{x}_{t+1})] + \frac{\gamma^2 L \sigma^2}{2}$$

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which can be rearranged as

$$\gamma \left(1 - \frac{\gamma L}{2}\right) \|\nabla F(\mathbf{x}_t)\|^2 \leq \mathbb{E}[F(\mathbf{x}_t) - F(\mathbf{x}_{t+1})] + \frac{\gamma^2 L \sigma^2}{2}$$

Note that $\gamma \leq \frac{1}{L}$, and hence $1 - \frac{\gamma L}{2} \geq \frac{1}{2}$.

$$\frac{\gamma}{2}\|\nabla F(\mathbf{x}_t)\|^2 \leq F(\mathbf{x}_t) - F(\mathbf{x}^*) - (\mathbb{E}[F(\mathbf{x}_{t+1}) - F(\mathbf{x}^*)]) + \frac{\gamma^2 L \sigma^2}{2}.$$

Proof of Theorem 3.

Taking expectation and telescoping, we get

$$\frac{\gamma}{2}\mathbb{E}\left[\sum_{t=1}^{T}\|\nabla F(\mathbf{x}_t)\|^2\right] \leq F(\mathbf{x}_1) - F(\mathbf{x}^*) - \left(\mathbb{E}[F(\mathbf{x}_{T+1}) - F(\mathbf{x}^*)]\right) + \frac{\gamma^2 L\sigma^2}{2}\cdot T$$
$$\leq F(\mathbf{x}_1) - F(\mathbf{x}^*) + \frac{\gamma^2 L\sigma^2}{2}\cdot T$$

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$$\leq F(\mathbf{x}_1) - F(\mathbf{x}^*) + \frac{\gamma^2 L\sigma^2}{2}\cdot T$$

And

$$\mathbb{E}\left[\sum_{t=1}^{T}\frac{1}{T}\|\nabla F(\mathbf{x}_t)\|^2\right] \leq \frac{2F(\mathbf{x}_1) - F(\mathbf{x}^*)}{T\gamma} + \gamma L\sigma^2$$

Since $\gamma := \min\{\frac{1}{L}, \frac{\gamma_0}{\sigma\sqrt{T}}\}$, $\gamma \leq \frac{\gamma_0}{\sigma\sqrt{T}}$ and $\frac{1}{r} = \max\{L, \frac{\sigma\sqrt{T}}{\gamma_0}\}$ $\square$

Proof of Theorem 3.

$$
\begin{aligned}
\mathbb{E}\left[\sum_{t=1}^{T}\frac{1}{T}\|\nabla F(\mathbf{x}_t)\|^2\right] &\le \frac{2F(\mathbf{x}_1)-F(\mathbf{x}^*)}{T}\cdot\max\{L,\frac{\sigma\sqrt{T}}{\gamma_0}\}+L\sigma^2\cdot\frac{\gamma_0}{\sigma\sqrt{T}} \\
&\le \frac{2L(F(\mathbf{x}_1)-F(\mathbf{x}^*))}{T}+\frac{\sigma}{\sqrt{T}}\left(\frac{2(F(\mathbf{x}_1)-F(\mathbf{x}^*))}{\gamma_0}+L\gamma_0\right)
\end{aligned}
$$

$\square$

Complexity

Table: Complexity comparison for GD and SGD, $\kappa = \frac{L}{\mu}$.

Problem class	Method	Iter. complexity	Iter. cost	Total
Strongly convex and smooth	GD	$\mathcal{O}(\kappa \log(1/\varepsilon))$	$\mathcal{O}(n)$	$\mathcal{O}(n\kappa \log(1/\varepsilon))$
	SGD	$\mathcal{O}(1/\varepsilon)$	$\mathcal{O}(1)$	$\mathcal{O}(1/\varepsilon)$
Non-convex	GD	$\mathcal{O}(1/\varepsilon^2)$	$\mathcal{O}(n)$	$\mathcal{O}(n/\varepsilon^2)$
	SGD	$\mathcal{O}(1/\varepsilon^4)$	$\mathcal{O}(1)$	$\mathcal{O}(1/\varepsilon^4)$